

* Landmarks in the Space of QFT's:

- QFT become scale-invariant at long distances

UV complete QFT: a renormalization group (RG) flow between CFTs

$$\left. \begin{array}{l} \text{(CFT)}_{UV} \\ \downarrow \\ \text{(CFT)}_{IR} \end{array} \right\} \text{QFT}$$

- Having a CFT in IR is generic
- Studying CFT lets us map out the possible endpoints of RG-flow
- Most (interesting) RG flows are nonperturbative.

ex: ϕ^4 in $D=3$

$$S = \int d^3x \left[-\frac{1}{2} (\partial\phi)^2 - \frac{1}{2} m^2 \phi^2 - \frac{1}{4!} g \phi^4 \right]$$

This theory is

(i) free in the UV since $[m] = [g] = 1$ and can be ignored at high energies.

(ii) $m^2 \gg 0$, IR theory is massive and preserves $\mathbb{Z}_2$ symmetry

(iii) $m^2 \ll 0$, the IR massive but spontaneously breaks $\mathbb{Z}_2$

(iv) $m^2 \rightarrow g^2/m^2$, IR becomes gapless and is described by a nontrivial interacting CFT.

• Study with $d \rightarrow d-\epsilon$, $\epsilon \rightarrow 1$

Let us study this theory:

$$\mathcal{L} = -\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 - \frac{1}{4!} g \phi^4$$

* A $\mathbb{Z}_2$ symmetry: leaves $\mathcal{L}$ invariant under $\phi \rightarrow -\phi$

$$Z^{-1} \phi(x) Z = -\phi(x) \quad | \quad Z^2 = 1 \quad Z^{-1} = Z = Z^\dagger$$

$$\mathcal{L} = -\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi)$$

$$V(\phi) = \frac{1}{2} m^2 \phi^2 + \frac{1}{24} g \phi^4$$

• As usual, we can expand $\phi \rightarrow \phi - v$, in anticipation

$$V(0) \Rightarrow \phi^2 \left(\frac{1}{2} m^2 + \frac{1}{24} g \phi^2 \right) = 0$$

$$\frac{\partial V}{\partial \phi} = (m^2 \phi + \frac{1}{6} g \phi^3) = 0$$

$$(m^2 + \frac{1}{6} g \phi^2) \phi = 0$$

$$\phi = 0$$

or

$$\phi^2 = -\frac{6m^2}{g}$$

→ Define: $V^2 \equiv +\sqrt{6|m^2|/g}$

$$V(\phi) = \frac{1}{24} g (\phi^2 - V^2)^2 - \frac{1}{24} g V^4$$

Constant: Can drop

$$V(\phi) = \frac{1}{24} g (\phi^2 - V^2)^2$$

• Two values which minimize the energy:

$$\begin{cases} \phi(x) = +V \\ \phi(x) = -V \end{cases}$$

→ Defined for $m^2 < 0$ otherwise $\mathbb{Z}_2$ is preserved and we only have $\phi = 0$

* Two Ground states

$$\langle +0 | \phi(x) | +0 \rangle = +V$$

$$\langle -0 | \phi(x) | -0 \rangle = -V$$

$$\langle 0 | \phi(x) | 0 \rangle = 0$$

$$\hookrightarrow \text{shift } \phi \rightarrow \phi \pm V$$

$$\star \langle 0 | \phi | 0 \rangle = \pm V$$

$$\mathbb{Z} | 0+ \rangle = | 0- \rangle : \langle +0 | -0 \rangle = 0$$

→ To motivate $\langle +0 | -0 \rangle = 0$, take an example:

$$H = \frac{1}{2} p^2 + \frac{1}{24} g (x^2 - V^2)^2$$

$$\psi_{\pm}(x) = \langle x | 0_{\pm} \rangle \sim \exp\left(-\omega (x \mp V)^2 / 2\right)$$

$$\omega = (g V^2 / 3)^{1/2}$$

• However the true ground state is a symmetric linear combination of these.

$\Rightarrow \phi$: ∞ set of oscillators, one for each pt in M^d
and each with $H_i \sim H$.

Coupled together by $(\nabla\phi)^2$ terms

$\Rightarrow$ There is a tunneling amplitude for each oscillator
 $|0+\rangle \rightarrow |0-\rangle$ all the oscillators have to tunnel
(Tunnel) m , $m = \# \text{ of oscillators} \rightarrow \infty$

and so in the limit of infinite volume $\langle 0+|0-\rangle = 0$

* Can use $|0+\rangle$ or $|0-\rangle$ as the ground state
choose $|0+\rangle$:

$$\rho(x) \equiv \phi(x) - v$$

Such that

$$\langle 0+|\rho(x)|0+\rangle = 0$$

$$V(\phi) = \frac{1}{24} g (\phi^2 - v^2)^2$$

$$= \frac{1}{24} g (\cancel{\rho^2 + 2\rho v + v^2} - v^2)^2$$

$$(\rho^2 + 2\rho v + v^2 - v^2)^2 = (\rho^2 + 2\rho v)^2$$

$$= \rho^4 + 4\rho^2 v^2 + 2\rho^3 v$$

$$\therefore V(\phi) = \frac{g}{6} v^2 \rho^2 + \frac{g}{12} v \rho^3 + \frac{g}{24} \rho^4$$

$$\hookrightarrow \text{Physical Mass } \frac{g}{6} v^2 = \frac{1}{2} m_{\text{phys}}^2 = |M^2|$$

$$m_{\text{phys}}^2 = 2 |M^2|$$

$\Rightarrow$ Theory is now $\phi^3 \oplus \phi^4$

$$\mathcal{L} = -\frac{1}{2} \partial^\mu \phi \partial_\mu \phi - \frac{1}{2} m_\phi^2 \phi^2 - \frac{g}{6} v \phi^3 - \frac{1}{24} g \phi^4$$

$\mathbb{Z}_2$: Spontaneously broken

$$\text{Renormalization: } \mathcal{L} = -\frac{1}{2} Z_\phi \partial^\mu \phi \partial_\mu \phi - \frac{1}{2} Z_m m^2 \phi^2 - Z_\lambda \lambda \phi^2$$

$m^2 > 0$: Z_i can absorb all ∞ 's. in $d \leq 4$

$m^2 < 0$: Need an extra Z factor for ϕ^3 term

• Need to add a term $\sim \phi$ to cancel tadpoles

Quantum Action:

$$\Gamma_{\text{YM}} = -\frac{1}{2} \int d^4x \tilde{\phi} \left(\frac{1}{2} \partial^\mu \phi \partial_\mu \phi + \frac{1}{2} m^2 \phi^2 + \frac{g}{6} v \phi^3 + \frac{g}{24} \phi^4 \right)$$

$\Gamma(\varphi) = \sum_{n=0}^{\infty} \frac{1}{n!} \int \frac{d^4 k_1}{(2\pi)^4} \frac{d^4 k_2}{(2\pi)^4} (2\pi)^4 \delta^4(k_1 + \dots + k_n) \times T_n(k_1, \dots, k_n) \tilde{\varphi}(k_1) \dots \tilde{\varphi}(k_n)$
 computed when $m^2 > 0$ has $\Gamma(\varphi)$ and V_n well defined and finite.

$V_n \Rightarrow n \neq 2m$ and so $\Gamma(\varphi)$ inherits $\mathbb{Z}_2$
 $m^2 < 0$: $\frac{\delta \Gamma}{\delta \varphi(x)} = -J(x)$

which has solutions $\sim \langle \varphi \rangle$

For $J=0$ (No source): $\varphi(x) = v$

$\Gamma(\varphi) = \int d^4 x \left[-\mathcal{L}(\varphi) - \frac{1}{2} \tilde{\mathcal{L}}(\varphi) \partial^\mu \varphi \partial_\mu \varphi + \dots \right]$

↑ loop corrected potential

$* \mathcal{L} = -\frac{1}{2} Z_\varphi \partial^\mu \varphi \partial_\mu \varphi - \frac{1}{2} Z_m m^2 \varphi^2 - \frac{1}{24} Z_\lambda g \varphi^4$

$d = 4 - \epsilon$, g : dimensionless

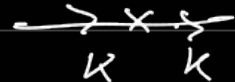
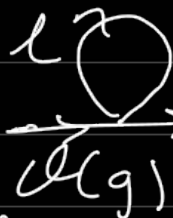
$\mathbb{Z}_2: \varphi \rightarrow -\varphi$

$m^2 > 0$

Regularize: $g \rightarrow g \tilde{\mu}^\epsilon$

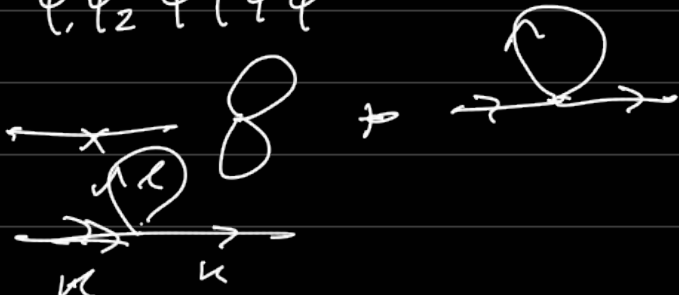
$[\mu] = [m]$

$\Pi(k^2)$: loop correction



For two point:

$\langle \varphi_1 \varphi_2 \rangle \sim \langle \varphi_1 \varphi_2 \rangle e^{\int \frac{1}{24} Z_\lambda g \varphi^4 d^4 x}$
 $\sim \varphi_1 \varphi_2 \sqrt{\varphi \varphi \varphi \varphi}$



$\sim \int \frac{d^4 k}{(2\pi)^4} \frac{d^4 p}{(2\pi)^4} \Delta(0) \Delta(x_1 - w) \Delta(x_2 - w)$
 $= \frac{i Z_\lambda g}{24} \int d^4 w \Delta(0) \int \frac{d^4 k}{(2\pi)^4} \frac{e^{-i(k_1 - w) \cdot x_1}}{k^2 + m^2} \int \frac{d^4 p}{(2\pi)^4} \frac{e^{-i(k_2 - w) \cdot x_2}}{p^2 + m^2}$
 $= \frac{i Z_\lambda g}{24} \Delta(0) \int \frac{d^4 k}{(2\pi)^4} \int \frac{d^4 p}{(2\pi)^4} \left(\frac{1}{k^2} \right) \left(\frac{1}{p^2} \right) e^{-i x_1 \cdot k} e^{-i x_2 \cdot p} \int d^4 w e^{i w \cdot (k + p)}$
 $= \frac{i Z_\lambda g \tilde{\mu}^2}{24} \Delta(0) \int \frac{d^4 k}{(2\pi)^4} \left(\frac{1}{k^2 + m^2} \right)^2 e^{-i p \cdot (x_1 - x_2)}$

$$i\Gamma(k^2) = \frac{1}{2} (-ig\hat{\mu}^\varepsilon) \left(\frac{i}{\varepsilon} \Delta(0) - i(Ak^2 + Bm^2) \right)$$

$$A = Z_p - 1 \quad \Delta(0) = \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{\ell^2 + m^2}$$

$$B = Z_m - 1$$

$$\Rightarrow \hat{\mu}^\varepsilon \Delta(0) = \frac{(-i)}{(4\pi)^2} \left[\frac{2}{\varepsilon} + 1 + \ln\left(\frac{\mu^2}{m^2}\right) \right] m^2$$

where

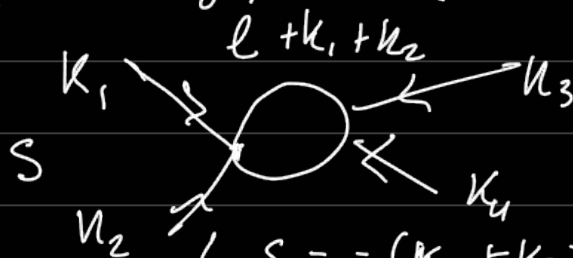
$$\mu \equiv \sqrt{4\pi} e^{-\gamma} \hat{\mu}^{\frac{\varepsilon}{2}}$$

$$\Rightarrow \Gamma(k^2) = \frac{g}{2(4\pi)^2} \left[\frac{2}{\varepsilon} + 1 + \ln\left(\frac{\mu^2}{m^2}\right) \right] m^2 - Ak^2 - Bm^2$$

$$A = \mathcal{O}(g^2)$$

$$B = \frac{g}{16\pi^2} \left(\frac{1}{\varepsilon} + K_B \right) + \mathcal{O}(g) \quad (\text{in } \overline{MS} \quad K_B = 0)$$

Vertex Correction:



$$+ k_2 \leftrightarrow k_3 + k_2 \leftrightarrow k_4$$

$$\begin{cases} S = -(k_1 + k_2)^2 =: S_1 \\ U = -(k_1 + k_3)^2 =: S_2 \\ Z = -(k_1 + k_4)^2 =: S_3 \end{cases}$$

$$iV_4(k_1, k_2, k_3, k_4) = -iZ_1 g + \frac{1}{2} (-ig)^2 \left(\frac{1}{i} \right)^2 \left[\sum_i F(-S_i) \right]$$

$$iF(k^2) = \hat{\mu}^\varepsilon \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{[(\ell+k)^2 + m^2](\ell^2 + m^2)} + \mathcal{O}(g^3)$$

$$= \frac{i}{16\pi^2} \left[\frac{2}{\varepsilon} + \int_0^1 dx \ln\left(\frac{\mu^2}{D}\right) \right]$$

$$D \equiv x(1-x)k^2 + m^2$$

$$\text{For } Z_1 = 1 + C$$

$$C = \frac{3g}{16\pi^2} \left(\frac{1}{\varepsilon} + K_C \right) + \mathcal{O}(g^2)$$

* The Beta function: $\beta(g) = \frac{dg}{d\ln(\mu)}$

taken with g_0 held fixed

$$g_0 = Z_g Z_\varphi^{-2} g \hat{\mu}^\varepsilon$$

$$\ln[Z_g Z_\varphi^{-2}] = \frac{3g}{16\pi^2} \frac{1}{\varepsilon} + \mathcal{O}(g^2)$$

$$\sqrt{m^2 > 0}$$

$$\beta(g) = g^2 L_1'(g) \Rightarrow \beta(g) = \frac{3g^2}{16\pi^2} + \mathcal{O}(g^3)$$

$\beta(g)$ is positive $\Rightarrow$ theory becomes more and more strongly coupled as $p \rightarrow \infty$

For $m^2 < 0$:

$$\mathcal{L} = -\frac{1}{2} \partial_\mu \rho \partial^\mu \rho - \frac{1}{2} \left(\frac{3}{2} Z_g - \frac{1}{2} Z_m \right) m_p^2 \rho^2 + \frac{1}{2} (Z_m - Z_g) \left(\frac{3}{g \tilde{\mu}^2} \right)^{1/2} m_p^3 \rho^3 - \frac{1}{6} Z_g (3g \tilde{\mu}^2)^{1/2} m_p \rho^3 - \frac{1}{24} Z_g g \tilde{\mu}^2 \rho^4$$

• Can now compute 1-loop corrections

(i) Start with vacuum loop corrections

$$\langle \phi \rangle = 0 \quad \text{at } \mathcal{O}(g)$$

$$\text{---} \times \text{---} + \text{---} \bigcirc \text{---}$$

from ρ

from ρ^3

$$\sim -\frac{i}{2} Z_g (3g \tilde{\mu}^2)^{1/2} m_p = -i Z_g g_3$$

$$g_3 = (3g \tilde{\mu}^2)^{1/2} m_p$$

• 1 pt. vertex factor $iY = i \left[\frac{1}{2} (Z_m - Z_g) \left(\frac{3}{g \tilde{\mu}^2} \right)^{1/2} m_p^3 \right]$

$$\langle \phi(x) \phi(0) \rangle = (iY + \frac{1}{2} (-i Z_g g_3) \frac{1}{i} \Delta(0)) \int d^4y \left(\frac{1}{i} \right) \Delta(x-y) + \mathcal{O}(g^2)$$

• Factor takes the form:

$$\text{For } \tilde{\mu}^{3/2} \Delta(0) \approx \frac{-i}{(4\pi)^2} \left(\frac{2}{(-\frac{\epsilon}{2})} + 1 + \ln\left(\frac{\mu^2}{m^2}\right) \right) m^2$$

$$\text{For } g_3 : \tilde{\mu}^{3/2} \Delta(0) \approx \frac{-i}{(4\pi)^2} \left(\frac{2}{(\frac{\epsilon}{2})} + 1 + \ln\left(\frac{\mu^2}{m^2}\right) \right) m^2$$

$$\Rightarrow \left(i \frac{1}{2} (Z_m - Z_g) \left(\frac{3}{g} \right)^{1/2} \left(\frac{-i}{(4\pi)^2} \left(\frac{2}{(-\frac{\epsilon}{2})} + 1 + \ln\left(\frac{\mu^2}{m^2}\right) \right) m^2 \right. \right.$$

$$\left. + \frac{1}{2} (-i Z_g (g_3)^{1/2} \left(\frac{-i}{(4\pi)^2} \left(\frac{2}{(-\frac{\epsilon}{2})} + 1 + \ln\left(\frac{\mu^2}{m^2}\right) \right) m^2 \right) \right)$$

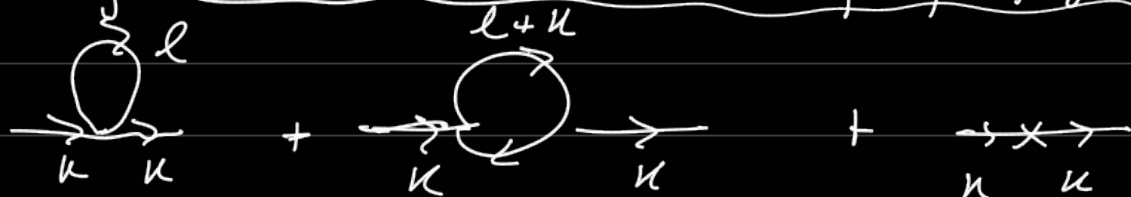
$$= \frac{i}{2} \left(\frac{3}{g} \right)^{1/2} m_p \left[Z_m - Z_g + \frac{g}{16\pi^2} \left(\frac{2}{\epsilon} + 1 + \ln\left(\frac{\mu^2}{m_p^2}\right) \right) + \mathcal{O}(g^2) \right]$$

$$\left[Z_m = 1 + B \quad \text{and} \quad Z_g = 1 + C \right]$$

$$\left[\frac{1}{\epsilon} = \frac{1}{r\pi^2} \right] k_0 - 3k_1 + 1 + \ln\left(\frac{\mu^2}{m^2}\right)$$

All $1/\epsilon$ factors cancel leaving
 $k_B - 3k_C = 1 - \ln(\mu^2/m_p^2)$

* $\mathcal{O}(g)$ corrections to the ρ -propagator $\langle \phi_1 \phi_2 \rangle$



$\sim \rho^4$ term $(\rho^3)^\epsilon$ -term ρ^2 term
 $(\phi_1 \phi \phi \phi \phi_2)$ $\phi_1 \phi \phi \phi \phi_2$

• Counter term $-i\mathcal{X}$

$$\mathcal{X} = A k^2 + \left(\frac{3}{2}C - \frac{1}{2}B\right)m_p^2$$

* Working in OS Scheme: $\Pi(-m_p^2) = 0$

$$\Pi(k^2) = \frac{1}{32\pi^2 m_p^2} \left[1 + \ln\left(\frac{\mu^2}{m_p^2}\right) + 3 \int_0^1 dx \ln(\mu^2/D) - (2k_C - k_B) \right]$$

$$= A k^2 + \mathcal{O}(k^2)$$

* Vertex terms:

$$V_3(k_1, k_2, k_3) = -g_3 \left(1 + C - \frac{3g}{16\pi^2} \frac{1}{\epsilon} + \mathcal{O}(g^2) \right)$$

$\Rightarrow$ divergent terms cancel to $\mathcal{O}(g)$

$\Rightarrow \mathbb{Z}_2$ -Symmetry is spontaneously broken for $m^2 < 0$

$\Rightarrow$ We get a UV-complete convergent theory which can be used to understand it as a CFT.

* Take the limit $g \rightarrow 1$
 $d - \epsilon \rightarrow d$

$$m_g^2 \rightarrow g^2/m^2: V_3 \sim (3g\hat{\mu})^{1/2} \sqrt{\frac{g}{m_p}} \left(1 + C - \frac{3g}{16\pi^2} + \dots \right)$$

$$V_3 \sim g(3\hat{\mu}) \frac{1}{m_p}$$

$\Rightarrow$ Non trivial CFT

Critical Universality:

IR CFTs can also arise from very different microscopic systems.

ex: 3D Ising Model: Lattice of classical spins

$S \in \{\pm 1\}$ with nearest neighbor interactions

$$Z_{\text{Ising}} = \sum_{\{S_i\}} \exp\left(-J \sum_{\langle ij \rangle} S_i S_j\right)$$

"discrete path-integral" over the map
 $S: \text{Lattice} \rightarrow \{\pm 1\}$

Same CFT as the ϕ^4 .

* ϕ^4 , Ising, water at critical pts, uni-axial magnets
are IR equivalent at their critical points.
"dualities"

Bootstrap Philosophy:

Critical universality: The ubiquity of IR equivalences
 $\Rightarrow$ We can study 3D Ising CFT by doing calculations in
any of its microscopic realizations.

- (0) Focus on the CFT itself and not a specific microscopic realization
- (1) Determine the full consequences of symmetries
- (2) Impose consistency conditions
- (3) Combine (1) and (2) to constrain or even solve the theory

ex: 6D $N=(2,0)$ SCFT describing the IR limit
L of a stack of M5 branes in M-theory

* Goal of Bootstrap: Eventually provide a fully
non-perturbative formulation of QFT: removing the
need for a Lagrangian.

Work in IE:

$$\partial_{\mu} T^{\mu\nu} = 0 \quad (\text{Operator Eq})$$

In the presence of other operators:

$$\partial_{\mu} \langle T^{\mu\nu}(x) \mathcal{O}_1(x_1) \dots \mathcal{O}_n(x_n) \rangle = - \sum_i \delta(x-x_i) \partial_i^{\nu} \langle \mathcal{O}_1(x_1) \dots \mathcal{O}_n(x_n) \rangle$$

* Consider a QFT over g - S[g, ϕ]

$$\langle \mathcal{O}_1 \dots \mathcal{O}_n \rangle_g = \int D\phi \mathcal{O}_1 \dots \mathcal{O}_n e^{-S[g, \phi]}$$

$$\langle T^{\mu\nu} \mathcal{O}_1 \dots \mathcal{O}_n \rangle_g = \frac{2}{g} \frac{\delta}{\delta g_{\mu\nu}(x)} (\langle \mathcal{O}_1 \dots \mathcal{O}_n \rangle)$$

* Introduce Source terms $\sum_i \int d^4x \sqrt{g} J_i \phi_i$

$$Z_J[\phi] = \int D\phi e^{iS + i \sum_i \int d^4x \sqrt{g} J_i \phi_i}$$

* $\delta g_{\mu\nu} = \nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu$, $\delta\phi = \xi^\lambda \partial_\lambda \phi$
 $S' = S$:

$$\phi \rightarrow \phi + \delta\phi$$

$$S' = S + \delta S$$

$$\delta S = -\frac{1}{2} \int d^4x \sqrt{g} T^{\mu\nu}(x) \delta g_{\mu\nu} = -\int d^4x \sqrt{g} T^{\mu\nu} \nabla_\mu \xi_\nu$$

$$\delta \left(\frac{1}{i} \frac{\partial}{\partial J_1} \dots \frac{1}{i} \frac{\partial}{\partial J_n} Z[J, \phi] \right) = 0$$

$$\int D\phi \delta(\phi_1 \dots \phi_n) e^{-iS + i \sum_i \int d^4x \sqrt{g} J_i \phi_i} + \int D\phi \phi_1 \dots \phi_n \delta S e^{-iS + i \sum_i \int d^4x \sqrt{g} J_i \phi_i} = 0$$

Set $J=0$: $\delta\phi_i = i \xi^\lambda \partial_\lambda \phi_i + (\dots)$

$$0 = \int D\phi \xi^\lambda \left(\sum_i \nabla_\lambda \phi_i + \dots \right) \phi_1 \dots \phi_{i-1} \phi_{i+1} \dots \phi_n e^{iS}$$

$$+ \int D\phi \phi_1 \dots \phi_n \int d^4x \sqrt{g} T^{\mu\nu} \nabla_\mu \xi_\nu$$

$$\xi_\lambda \sum_i [\nabla_\lambda]^\lambda \delta(x-x_i) \langle \phi_1 \dots \phi_n \rangle = \xi_\lambda \nabla_\mu \langle T^{\mu\lambda} \phi_1 \dots \phi_n \rangle$$

* Word-Takeaway:

$$\nabla_\mu \langle T^{\mu\nu} \phi_1 \dots \phi_n \rangle = - \sum_i \delta(x-x_i) \langle \phi_1 \dots \phi_{i-1} (\xi^\lambda \partial_\lambda \phi_i) \phi_{i+1} \dots \phi_n \rangle$$

* $P^\nu(\Sigma) \equiv - \int_\Sigma dS_\mu T^{\mu\nu}(x)$ (Boundary surface Σ)

* Word identity implies $\langle P^\nu \phi_1 \dots \phi_n \rangle$ does not change as we move Σ , as long as Σ doesn't cross any operator insertions

$P^\nu(\Sigma)$: Topological Surface operator



$$\langle P^\nu \phi_1 \dots \phi_n \rangle = - \left\langle \int dS_\mu T^{\mu\nu}(x) \phi_1 \dots \phi_n \right\rangle$$

$$\Rightarrow \nabla_\mu \langle P^\nu \phi_1 \dots \phi_n \rangle = 0$$

so long as $x \neq x_i$: if this occurs then $\nabla_\mu \mathcal{O}(x) \neq 0$ and the correlator $\langle P^\mu \mathcal{O}_1 \dots \mathcal{O}_n \rangle$ will no longer be unchanged.

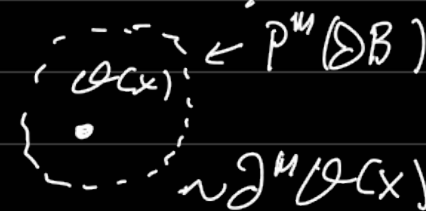
- Let $\Sigma = \partial B$ be the Boundary of a ball B containing x and no other insertions

$$\int_B \partial_\mu \langle T^{\mu\nu}(x) \mathcal{O}(x) \dots \mathcal{O}_n \rangle = - \int_B \delta(x-x_i) \Sigma^\mu \partial_\mu \langle \mathcal{O}(x) \dots \rangle$$

$$* \int_B d\omega = \int_{\partial B} \omega$$

$$\int_{\partial B} dS_\mu \langle T^{\mu\nu} \mathcal{O}(x) \dots \mathcal{O}_n \rangle = - \partial^\nu \langle \mathcal{O}(x) \dots \rangle$$

$$\langle P^\mu(\Sigma) \mathcal{O}(x) \dots \rangle = \partial^\mu \langle \mathcal{O}(x) \dots \rangle$$

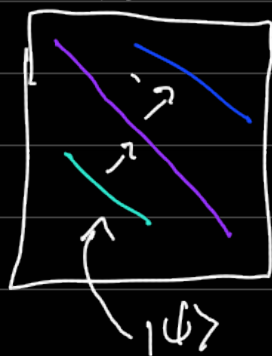


- Surrounding $\mathcal{O}(x)$ with P^μ gives a derivative.

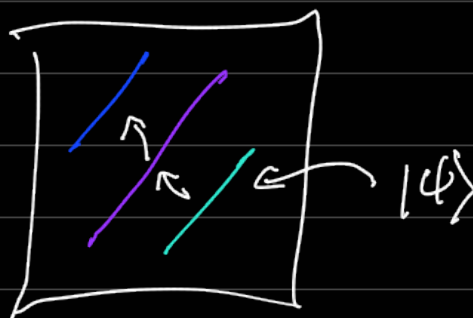
* Having a topological condim 1 operator is the same as having a symmetry

Here P^μ is of course just the conserved charge

ex: In a rotationally invariant theory in $\mathbb{R}^d$ we can choose any direction as time and states live orthogonal to this direction: states $|\psi\rangle$ live on slices



or



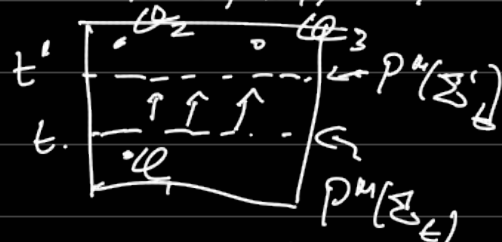
- Each interpretation is a "quantization"

- To specify a Quantization, we foliate spacetime by slices related by an isometry ∂_t

$$\langle \mathcal{O}_1(x_1) \dots \mathcal{O}_n(x_n) \rangle = \langle 0 | T \{ \hat{\mathcal{O}}_1(t_1, x_1) \dots \hat{\mathcal{O}}_n(t_n, x_n) \} | 0 \rangle$$

Topological Operators & Symmetries:

$P^\mu(\Sigma_t)$: Fact that P^μ is Topological is the path integral version



of the statement that momenta is conserved.

- Understand what happens when we move P^μ past an operator insertion.

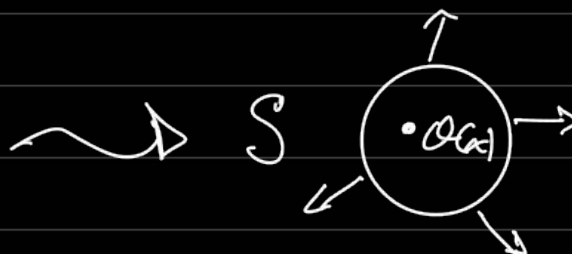
$$\mathcal{O}(x): t, \Sigma_1: t_1 \text{ to } \Sigma_2: t_2$$

$$t_1 < t < t_2$$

$$\uparrow \quad \uparrow \quad \text{---} \quad \Sigma_2$$

$$\bullet \mathcal{O}(x)$$

$$\text{---} \quad \downarrow \quad \downarrow \quad \text{---} \quad \Sigma_1$$



- When we quantize our theory the difference $P^\mu(\Sigma_2) - P^\mu(\Sigma_1)$ due to time ordering

$$\begin{aligned} \langle P^\mu(\Sigma_2) - P^\mu(\Sigma_1) \mathcal{O}(x) \dots \rangle &= \langle 0 | (\hat{P}^\mu(\Sigma_2) \hat{\mathcal{O}}(x) \dots - \hat{\mathcal{O}}(x) \hat{P}^\mu) | 0 \rangle \\ &= \langle 0 | T \{ [\hat{P}^\mu, \hat{\mathcal{O}}(x)] \dots \} | 0 \rangle \end{aligned}$$

- Because P^μ is Topological, we can deform

$$\Sigma_2 - \Sigma_1 \cong S^{D-1} \text{ surrounding } \mathcal{O}(x)$$

$$\Rightarrow \text{For any charge } Q(\Sigma): Q(\Sigma_2) - Q(\Sigma_1) = Q(\Sigma_2 - \Sigma_1) \xrightarrow{\text{deform}} Q(S)$$

$$\begin{aligned} \langle 0 | T \{ [\hat{P}^\mu, \hat{\mathcal{O}}(x)] \dots \} | 0 \rangle &= \langle (P^\mu(\Sigma_2) - P^\mu(\Sigma_1)) \mathcal{O}(x) \dots \rangle \\ &= \langle P^\mu(S) \mathcal{O}(x) \dots \rangle \\ &= \partial_\mu \langle \mathcal{O}(x) \dots \rangle \\ &= \partial_\mu \langle 0 | T \{ \hat{\mathcal{O}}(x) \dots \} | 0 \rangle \end{aligned}$$

$$[\hat{P}^\mu, \hat{\mathcal{O}}(x)] = \partial^\mu \hat{\mathcal{O}}(x)$$

- Independent of how we quantize the theory
- $$[P^\mu, \mathcal{O}(x)] = \partial^\mu \mathcal{O}(x)$$

without specifying a quantization.

$\mathcal{O}(x)$: Path integral insertions

$\hat{\mathcal{O}}(x)$: Quantum operators

$$[Q, \mathcal{O}(x)]:$$

is Actual Commutator $[\hat{Q}, \hat{\mathcal{O}}]$ in any Quant.

is Surrounding $\mathcal{O}(x)$ with a topological surface operator $Q(S)$.

* Integrating $[P^\mu, \mathcal{O}(x)] = \partial^\mu \mathcal{O}(x)$ we find

$$\mathcal{U}(x) = e^{xP} \mathcal{U}(0) e^{-xP}$$

• In Path-integral $e^{xP}(\Sigma)$ is another topological surface operator.

• When we surround $\mathcal{U}(0)$ with $e^{xP}(\Sigma)$ it becomes conjugation:

$$e^{xP}(\Sigma) \mathcal{U}(0) \rightarrow e^{\hat{P}x} \hat{\mathcal{U}}(0) e^{-\hat{P}x}$$

$$[e^{xP}(\Sigma), \mathcal{U}(0)] = \langle e^{xP}(\Sigma) \mathcal{U}(0) \dots \rangle = \sum_n \frac{1}{n!} \langle (xP)^n \mathcal{U}(0) \dots \rangle$$

$$\langle \mathcal{O}_1(x_1) \dots \mathcal{O}_n(x_n) \rangle = \langle 0 | e^{t_n P^0} \mathcal{O}_n(0, \vec{x}_n) e^{-t_n P^0} \dots e^{t_1 P^0} \mathcal{O}_1(0, \vec{x}_1) e^{-t_1 P^0} | 0 \rangle$$

$$t_n > t_{n-1} > \dots > t_1$$

P^0 will annihilate $|0\rangle$: $P^0 |0\rangle = \mathcal{L}(P^0)^\dagger = 0$

$$\langle \mathcal{O}_1(x_1) \dots \mathcal{O}_n(x_n) \rangle = \langle \mathcal{O}_n(0, \vec{x}_n) e^{-(t_n - t_{n-1}) P^0} \dots e^{-(t_2 - t_1) P^0} \mathcal{O}_1(0, \vec{x}_1) | 0 \rangle$$

* Path integral between spatial slices separated by time t computes the time evolution operator

$$U(t) = e^{-tP^0}$$

More Symmetries: $\partial_\mu \tilde{J}^\mu = 0$

Given $\tilde{J}^\mu$ we can always define a topological surface operator by integration.

* Given $E = E^\mu(x) \partial_\mu$, the charge

$$Q_E(\Sigma) = - \int_\Sigma dS_\mu E_\nu(x) T^{\mu\nu}(x)$$

• will be conserved whenever

$$0 = \partial_\mu (E_\nu T^{\mu\nu})$$

$$= \partial_\mu E_\nu T^{\mu\nu} + E_\nu \underbrace{\partial_\mu T^{\mu\nu}}_{=0}$$

$$= \frac{1}{2} (\partial_\mu E_\nu + \partial_\nu E_\mu) T^{\mu\nu} \Rightarrow \boxed{\partial_\mu E_\nu + \partial_\nu E_\mu = 0} \quad (\text{Killing Equation})$$

$$\mathcal{M}^{\mu\nu} (\partial_\mu E_\nu + \partial_\nu E_\mu) = 2 \partial^\mu E_\mu = 0$$

$$\leadsto \mathcal{M}_{\mu\nu} = \text{diag}(\underbrace{-1, \dots, -1}_s, \underbrace{+1, \dots, +1}_t)$$

$$ds'^2 = dx'^\mu dx'^\nu \mathcal{M}_{\mu\nu} \equiv \Omega(x) ds^2$$

$$x^{\mu} = f^{\mu}(x) \Rightarrow \partial_{\rho} f^{\mu} \partial_{\kappa} f^{\nu} \eta_{\mu\nu} = \eta_{\rho\kappa} \Omega$$

$$\partial^{\lambda} f^{\mu} \partial_{\kappa} f_{\mu} = \delta_{\kappa}^{\lambda} \Omega$$

$$\Omega = \frac{1}{D} \partial_{\rho} f^{\mu} \partial^{\rho} f_{\mu}$$

* Diffeomorphism: $f^{\mu}(x) = x^{\mu} + \epsilon^{\mu}(x)$

$$\frac{1}{D} \partial_{\rho} (x^{\mu} + \epsilon^{\mu}) \partial^{\rho} (x_{\mu} + \epsilon_{\mu}) = (\delta_{\rho}^{\mu} + \partial_{\rho} \epsilon^{\mu}) (\delta_{\mu}^{\rho} + \partial^{\rho} \epsilon_{\mu})$$

$$\delta_{\rho}^{\mu} \delta_{\mu}^{\rho} + \delta_{\rho}^{\mu} \partial^{\rho} \epsilon_{\mu} + \delta_{\mu}^{\rho} \partial_{\rho} \epsilon^{\mu} + \partial(\epsilon^2) = D \Omega(x)$$

$$(D + \partial^{\mu} \epsilon_{\mu} + \partial^{\mu} \epsilon_{\mu}) = D \Omega$$

$$2 \partial^{\mu} \epsilon_{\mu} = D(\Omega - 1) \Rightarrow \partial^{\mu} \epsilon_{\mu} = \frac{1}{2} D(\Omega - 1)$$

To lowest order:

$$\hookrightarrow \frac{2}{D} \partial^{\mu} \epsilon_{\mu} = (\Omega - 1)$$

$$\partial_{\rho} (x^{\mu} + \epsilon^{\mu}) \partial_{\kappa} (x_{\mu} + \epsilon_{\mu}) = \eta_{\rho\kappa} \Omega$$

$$(\delta_{\rho}^{\mu} + \partial_{\rho} \epsilon^{\mu}) (\eta_{\mu\kappa} + \partial_{\kappa} \epsilon_{\mu}) = \eta_{\rho\kappa} \Omega$$

$$\eta_{\rho\kappa} + \partial_{\kappa} \epsilon_{\rho} + \partial_{\rho} \epsilon_{\kappa} = \eta_{\rho\kappa} \Omega$$

$$\partial_{\kappa} \epsilon_{\rho} + \partial_{\rho} \epsilon_{\kappa} = \eta_{\rho\kappa} (\Omega - 1)$$

$$\partial_{\kappa} \epsilon_{\rho} + \partial_{\rho} \epsilon_{\kappa} = \eta_{\rho\kappa} \left(\frac{2}{D} \right) \partial^{\mu} \epsilon_{\mu}$$

(i) Take ∂^{ν} :

$$\partial^2 \epsilon_{\rho} + \partial_{\rho} \partial^{\mu} \epsilon_{\mu} = \left(\frac{2}{D} \right) \partial_{\rho} (\partial^{\mu} \epsilon_{\mu})$$

$$D \partial^2 \epsilon_{\rho} = (2 - D) \partial_{\rho} (\partial^{\mu} \epsilon_{\mu})$$

(ii) Act with ∂^{ρ}

$$D \partial^2 \partial^{\rho} \epsilon_{\rho} = (2 - D) \partial^2 (\partial^{\rho} \epsilon_{\rho})$$

$$\bullet 2(D - 1) \partial^2 \partial^{\rho} \epsilon_{\rho} = 0 \Rightarrow \text{For } (D \neq 1): \partial^2 \partial^{\rho} \epsilon_{\rho} = 0$$

$$\bullet D \neq 2: (2 - D) \partial_{\mu} \partial^{\kappa} \epsilon_{\kappa} = D \partial^2 \epsilon_{\mu}$$

$$(2 - D) \partial_{\rho} \partial_{\mu} \partial^{\kappa} \epsilon_{\kappa} = D \partial^2 \partial_{\rho} \epsilon_{\mu}$$

$$\partial_{\rho} \partial_{\mu} \partial^{\kappa} \epsilon_{\kappa} = \left(\frac{D}{2-D} \right) \partial_{\rho} \partial^2 \epsilon_{\mu}$$

$$\partial_{\rho} \partial_{\mu} \partial^{\kappa} \epsilon_{\kappa} = \left(\frac{1}{2-D} \right) \eta_{\rho\mu} \partial^2 \partial^{\kappa} \epsilon_{\kappa} = 0 \quad (D \neq 1)$$

$$\bullet (2 - D) \partial_{\mu} \eta^{\kappa\lambda} \partial_{\lambda} \epsilon_{\kappa} = D \eta^{\kappa\lambda} \partial_{\kappa} \partial_{\lambda} \epsilon_{\mu}$$

$$\text{or } (2 - D) \partial_{\mu} \partial_{\lambda} \epsilon_{\mu} = D \partial_{\kappa} \partial_{\lambda} \epsilon_{\mu}$$

$$\hookrightarrow \partial_{\rho} \partial_{\kappa} \partial_{\mu} \epsilon_{\nu} = -\partial_{\rho} \partial_{\kappa} \partial_{\nu} \epsilon_{\mu} + \frac{2}{D} \eta_{\mu\nu} \partial_{\rho} \partial_{\kappa} \partial^{\rho} \epsilon_{\rho}$$

$$= -\partial_{\rho} \partial_{\kappa} \partial_{\nu} \epsilon_{\mu} = \partial_{\rho} \partial_{\nu} \partial_{\mu} \epsilon_{\kappa}$$

$$= \partial_{\rho} \partial_{\mu} \partial_{\nu} \epsilon_{\kappa} = -\partial_{\rho} \partial_{\mu} \partial_{\kappa} \epsilon_{\nu}$$

$$= 0$$

$$\text{i.e. } \partial_{\rho} \partial_{\kappa} \partial_{\mu} \epsilon_{\nu} = 0$$

$$x^{\mu} = x^{\mu} + \epsilon^{\mu} = f^{\mu}$$

$$| \epsilon^{\mu} = a^{\mu} + t^{\mu}_{\nu} x^{\nu} + t^{\mu}_{\nu\rho} x^{\nu} x^{\rho}$$

$$x'^{\mu} = x^{\mu} + a^{\mu} + t^{\mu}_{\nu} x^{\nu} + t^{\mu}_{\nu\rho} x^{\nu} x^{\rho}$$

$$\begin{cases} t_{\mu\nu} + t_{\nu\mu} = \frac{2}{D} \eta_{\mu\nu} (t^{\mu}_{\mu}) \end{cases} \quad (a)$$

$$\begin{cases} t_{\mu\nu\rho} + t_{\nu\rho\mu} = \frac{2}{D} \eta_{\mu\nu} (t^{\mu}_{\rho}) \end{cases} \quad (b)$$

$$(a) \quad t_{\mu\nu} = \omega_{[\mu\nu]} + \lambda \eta_{\mu\nu}$$

$$t^{\mu}_{\mu} = \lambda D \Rightarrow \lambda = T^{\mu}_{\mu} / D$$

$$(b) \quad t_{\mu\nu\rho} = \frac{1}{D} (\eta_{\mu\nu} t^{\mu}_{\rho} + \eta_{\nu\rho} t^{\mu}_{\mu} - \eta_{\rho\mu} t^{\mu}_{\nu})$$

$$t^{\mu}_{\nu\rho} = \frac{1}{D} (\delta^{\mu}_{\nu} t^{\mu}_{\rho} + \delta^{\mu}_{\rho} t^{\mu}_{\nu} - \eta_{\rho\nu} t^{\mu}_{\mu})$$

$$\therefore t^{\mu}_{\nu\rho} x^{\nu} x^{\rho} = \frac{1}{D} (2 t^{\mu}_{\rho} x^{\rho} x^{\mu} - x^2 t^{\mu}_{\mu})$$

$$\text{Define } \lambda^{\mu} \equiv -\frac{1}{D} t^{\mu}_{\mu}$$

$$t^{\mu}_{\nu\rho} x^{\nu} x^{\rho} = -2 \lambda_{\rho} x^{\rho} x^{\mu} + x^2 \lambda^{\mu}$$

$$x'^{\mu} = x^{\mu} + a^{\mu} + \omega^{\mu}_{\nu} x^{\nu} + \lambda x^{\mu} + \lambda^{\mu} x^{\nu} x_{\nu} - 2 \lambda_{\rho} x^{\rho} x^{\mu}$$

$\uparrow$ translations $\uparrow$ Lorentz $\uparrow$ Scale (Dilatations) $\swarrow$ Special Conformal trans-l. (λ^{μ})

$$D + \frac{D(D-1)}{2} + 1 + D = \frac{1}{2} (D+1)(D+2) = \dim(\text{Conformal group})$$

$$\lambda^{\mu} = -\frac{2}{D} t^{\mu}_{\mu}$$

$$\text{Translation: } a^{\mu} e^{a^{\sigma} P_{\sigma}} x^{\mu} = x^{\mu} + a^{\sigma} (P_{\sigma} x^{\mu})$$

$$P_{\sigma} = \partial_{\sigma}$$

$$\text{Lorentz: } \omega_{\sigma\lambda} e^{\omega_{\sigma\lambda} M^{\sigma\lambda}} x^{\mu} = x^{\mu} + \omega_{\sigma\lambda} M^{\sigma\lambda} x^{\mu}$$

$$\Rightarrow \omega_{\sigma\lambda} M^{\sigma\lambda} x^{\mu} = \omega^{\mu}_{\lambda} x^{\lambda}$$

$$\omega_{\lambda\sigma} x^{\sigma} \partial^{\lambda} (x^{\mu}) = \omega_{\lambda\sigma} x^{\sigma} \eta^{\lambda\mu} = \omega_{\sigma}^{\mu} x^{\sigma}$$

$$\therefore \omega_{\sigma\lambda} M^{\sigma\lambda} = \omega_{\sigma\lambda} (x^{\lambda} \partial^{\sigma})$$

$$= \frac{1}{2} \omega_{\sigma\lambda} (x^{\lambda} \partial^{\sigma}) + \frac{1}{2} \omega_{\lambda\sigma} x^{\sigma} \partial^{\lambda}$$

$$= \omega_{\sigma\lambda} \left[\frac{1}{2} (x^{\lambda} \partial^{\sigma} - x^{\sigma} \partial^{\lambda}) \right]$$

$$J^{\mu\nu} = (x^{\mu} \partial^{\nu} - x^{\nu} \partial^{\mu}) - \frac{1}{2} \omega_{\sigma\lambda} J^{\mu\nu}$$

$$\text{Dilatation: } \lambda$$

$$x'^{\mu} = x^{\mu} + \lambda x^{\mu}$$

$$\lambda D \equiv \lambda x^{\sigma} \partial_{\sigma}$$

$$D = x^\sigma \partial_\sigma$$

$$\lambda^\mu$$

* SCT: λ^μ

$$e^{\lambda^\mu K_\mu} x^\mu = x^\mu + \lambda^\mu (x^\sigma x_\sigma) - 2 \lambda_\sigma x^\sigma x^\mu$$

$$= x^\mu + x_\sigma (\lambda^\mu x^\sigma - 2 \lambda^\sigma x^\mu)$$

$$\bullet x_\sigma x^\sigma \lambda^\mu = \lambda^\mu x^2 \partial_\mu$$

$$\bullet \lambda^\mu x^\sigma \partial_\sigma (x_\mu \bullet)$$

$$\lambda^\lambda x^\sigma \partial_\sigma (x_\lambda x^\mu)$$

$$= \lambda^\lambda x^\sigma (\eta_{\sigma\lambda} x^\mu + x_\lambda \delta_\sigma^\mu)$$

$$= \lambda_\sigma x^\sigma x^\mu + \lambda^\lambda x^\mu x_\lambda$$

$$= 2 x^\mu \lambda^\sigma x_\sigma$$

•

$$K_\mu = x^2 \partial_\mu - 2 x^\sigma x_\mu \partial_\sigma$$

$$\lambda^\nu K_\nu (x^\mu) = x^2 \lambda^\mu - 2 x^\sigma x_\nu \delta_\sigma^\mu \lambda^\nu$$

$$= x^2 \lambda^\mu - 2 x^\mu x_\nu \lambda^\nu$$

CFT Algebra:

$$[D, K_\mu] f = x^\lambda \partial_\lambda (x^2 \partial_\mu f - 2 x^\sigma x_\mu \partial_\sigma f) - (x^2 \partial_\mu - 2 x^\sigma x_\mu \partial_\sigma) (x^\lambda \partial_\lambda f)$$

$$\textcircled{a} = \cancel{x^\lambda} 2 x_\lambda \partial_\mu f + \lambda^\lambda x^2 \partial_\lambda \partial_\mu f - 2 x^\lambda (x_\mu \partial_\lambda f + x^\sigma \eta_{\mu\lambda} \partial_\sigma f + x^\sigma x_\mu \partial_\lambda \partial_\sigma f)$$

$$= \underline{2 x^2 \partial_\mu f} + \underline{x^2 \lambda^\lambda \partial_\lambda \partial_\mu f}$$

$$- 4 x_\mu x^\sigma \partial_\sigma f - 2 x_\mu x^\lambda x^\sigma \partial_\lambda \partial_\sigma f$$

$$\textcircled{b} \underline{x^2 \partial_\mu f} + \underline{x^2 x^\lambda \partial_\lambda \partial_\mu f}$$

$$- \underline{2 x_\mu x^\lambda \partial_\lambda f} - \underline{2 x_\mu x^\sigma \partial_\sigma \partial_\lambda f x^\lambda}$$

$$\textcircled{c} \textcircled{a} - \textcircled{b} = x^2 \partial_\mu f - 2 x_\mu x^\sigma \partial_\sigma f$$

$$= (x^2 \partial_\mu - 2 x_\mu x^\sigma \partial_\sigma) f$$

$$= K_\mu f$$

$$\therefore [D, K_\mu] = K_\mu$$

$$\bullet [D, P_\mu] = (x^\sigma \partial_\sigma \partial_\mu - (\partial_\mu + x^\sigma \partial_\sigma \partial_\mu))$$

$$= -P_\mu$$

Full conformal Algebra:

$$[D, P] = -P$$

$$[D, D] = -D$$

$$[K_m, J_{\mu\nu}] = \eta_{\mu m} K_\nu - \eta_{\nu m} K_\mu, \quad [K_m, K_n] = 0$$

$$[K_m, P_\nu] = -2\eta_{m\nu} D + 2J_{m\nu}$$

$$(S, r) : (-1, \dots, -1, +1, \dots, +1)$$

$\quad \quad \quad r \quad \quad \quad S$

Isomorphic to $SO(r+1, s+1)$

• For SET:

$$x'^\mu = \lambda^\mu x^\sigma x_\sigma - 2\lambda_\sigma x^\sigma x^\mu$$

$$x'^\mu x'_\mu = (x'')^2$$

$$\Rightarrow (\lambda^\mu x^2 - 2(\lambda x) x^\mu) (\lambda_\mu x^2 - 2(\lambda x) x_\mu)$$

$$= \lambda^2 x^2 x^2 - 4(\lambda x)(\lambda x) x^2 + 4(\lambda x)(\lambda x) x^2$$

